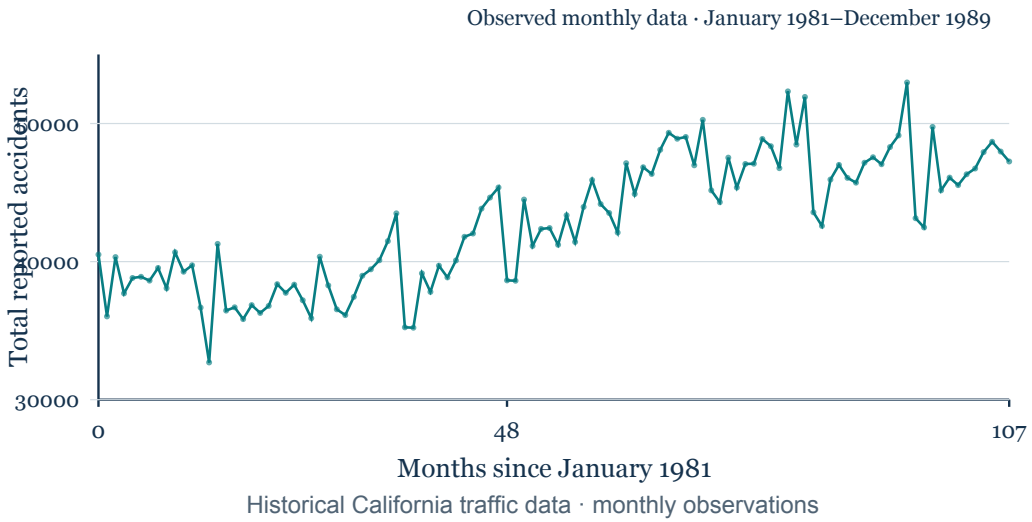
# Why can the same estimate have different uncertainty?

Monthly observations may share information; uncertainty calculations must account for it.




## Practice

Name a factor that might affect accidents for several consecutive months but is not fully measured in the dataset.

**Explanation:** A lasting change in road conditions, enforcement or reporting could affect several months. Such a shared influence can make the unexplained parts of their accident totals related.

# Define the monthly accident model

State the outcome, law indicators, calendar controls and error term.

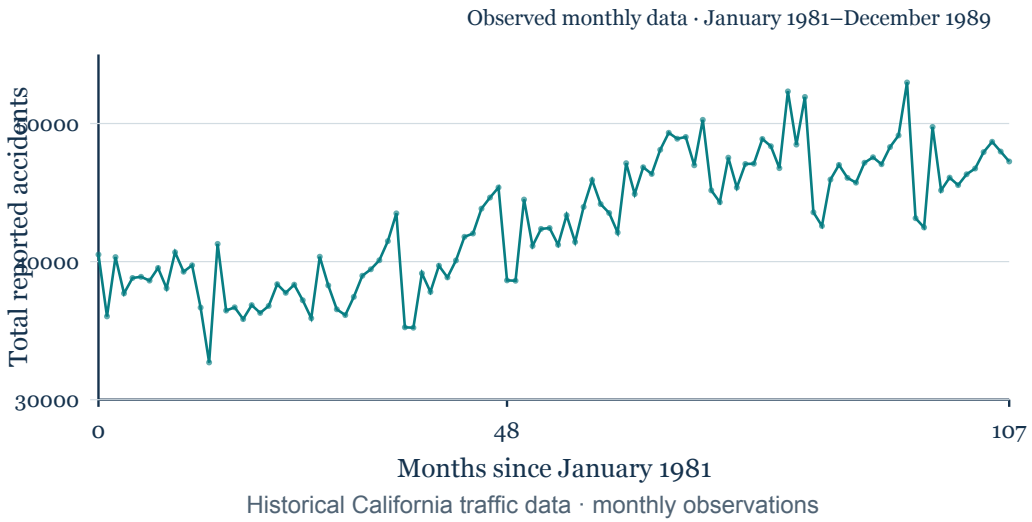
> $\log(A_t) = a + bB_t + cS_t + dU_t + fW_t + gt + \sum_{m=2}^{12} \gamma_m D_{mt} + u_t$

> $B, S$: law indicators; $U$: unemployment; $W$: weekend count.

> $D_{mt}$: month indicator; $u_t$: error.

# Separate the outcome pattern from the error pattern

Serial correlation concerns errors across dates, after specifying the model.



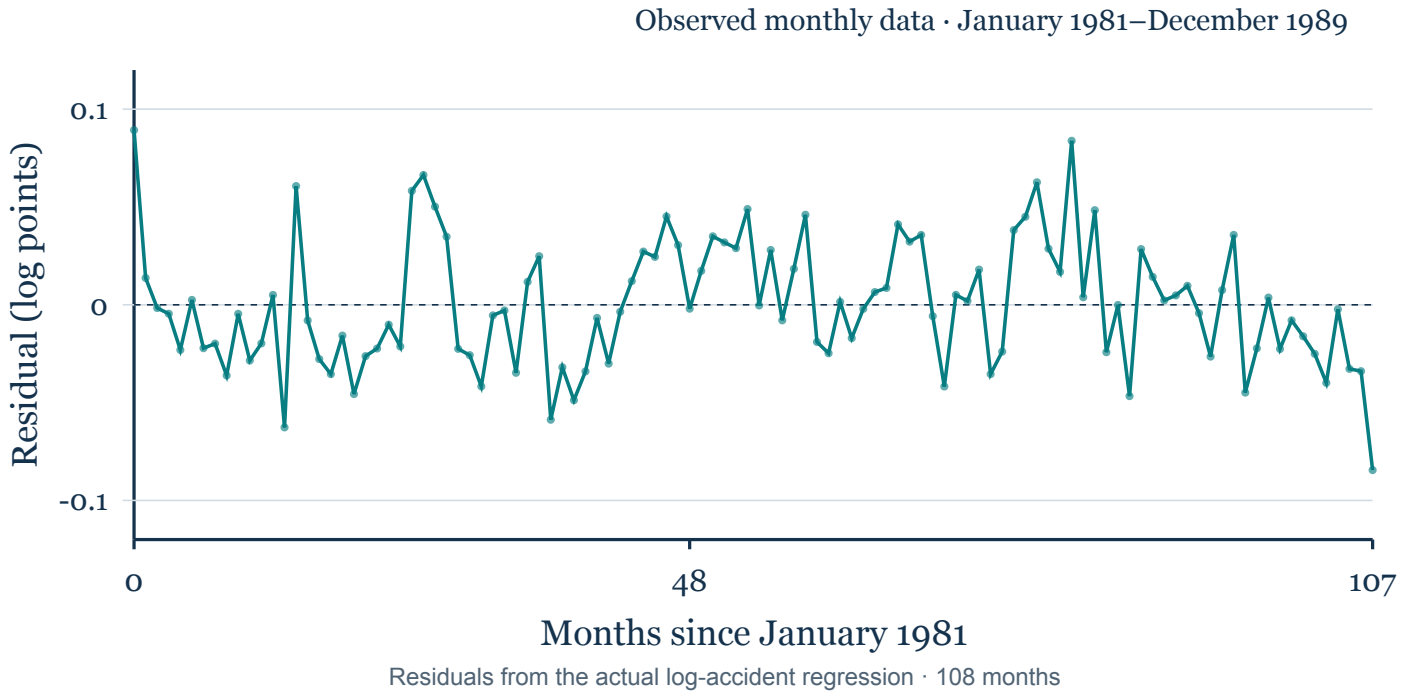

## Practice

Does a seasonal accident plot prove that the errors in a seasonally adjusted regression are serially correlated?

**Explanation:** No. Outcome seasonality can be explained by the included month effects. Residual dependence must be assessed after fitting the specified mean model.

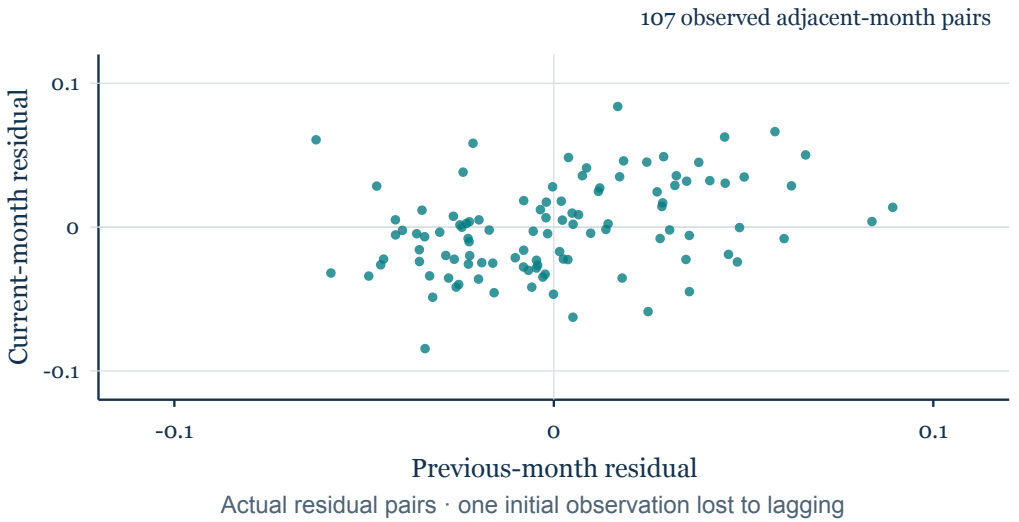
# Follow the residual path

Look for sequences of positive or negative residuals.



# Compare one month with the previous month

Put the previous month's residual on the horizontal axis.



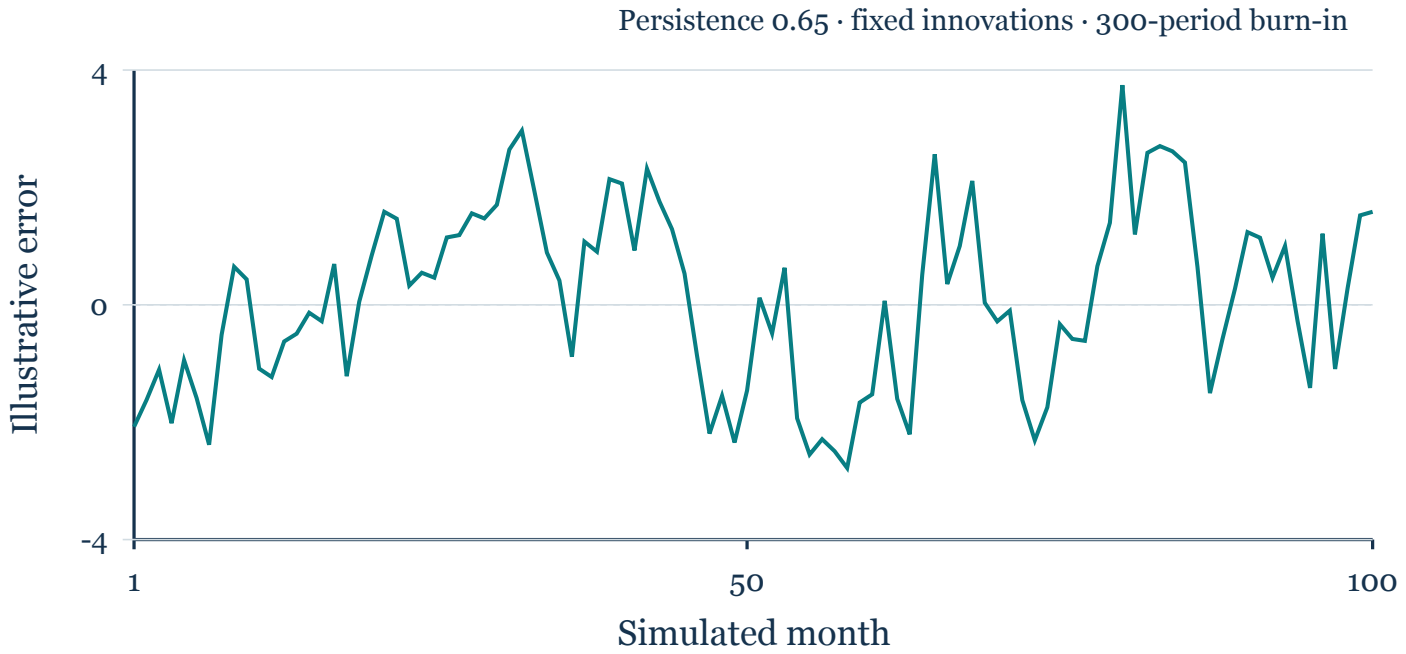

## Practice

Why are there 107 residual pairs? What does a point in the upper-right region mean?

**Explanation:** The first residual has no observed predecessor. An upper-right point is a month with a positive residual preceded by another month with a positive residual.

# See how an error can carry into the next month

An AR(1) process adds a new shock to a fraction of the previous error.



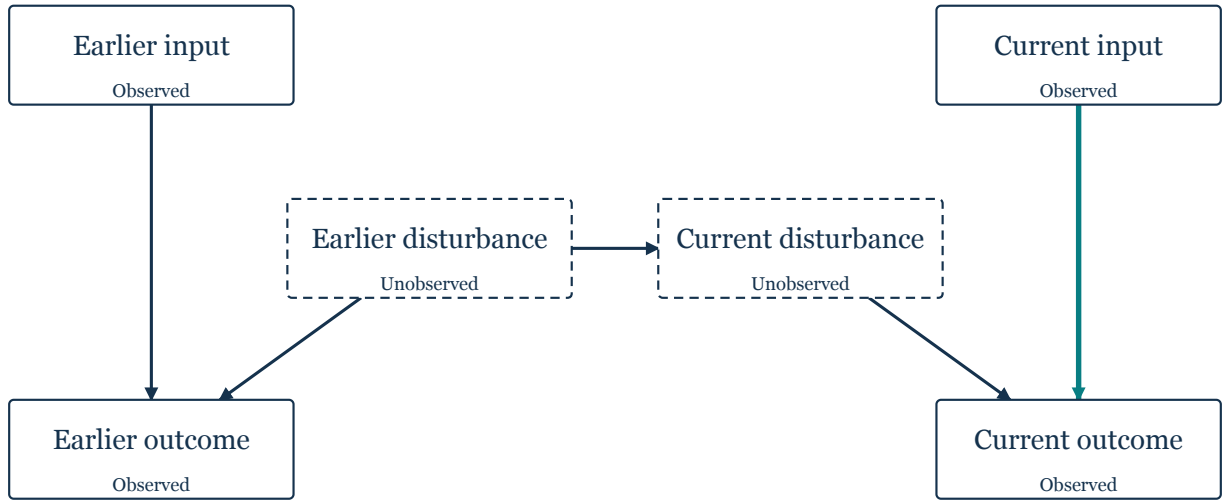

Illustrative simulation · fixed innovations · vertical range adapts to the selected persistence

$u_t = \rho u_{t-1} + \varepsilon_t$

$\rho$: persistence; $\varepsilon_t$: new unpredictable error

# Separate the comparison, model and uncertainty

Related errors do not, by themselves, tell us whether the coefficient is valid.




1 open directed route · 0 open alternatives · gold: conditioned on

Assumed causal structure · dashed boxes: unobserved · gold boxes: conditioned on

- Exogeneity: required lack of relation between regressors and error
- Mean model: which variables predict the outcome?
- SE calculation: how much uncertainty remains?

# Why dependence changes uncertainty

Related errors add covariance terms to the variance calculation.

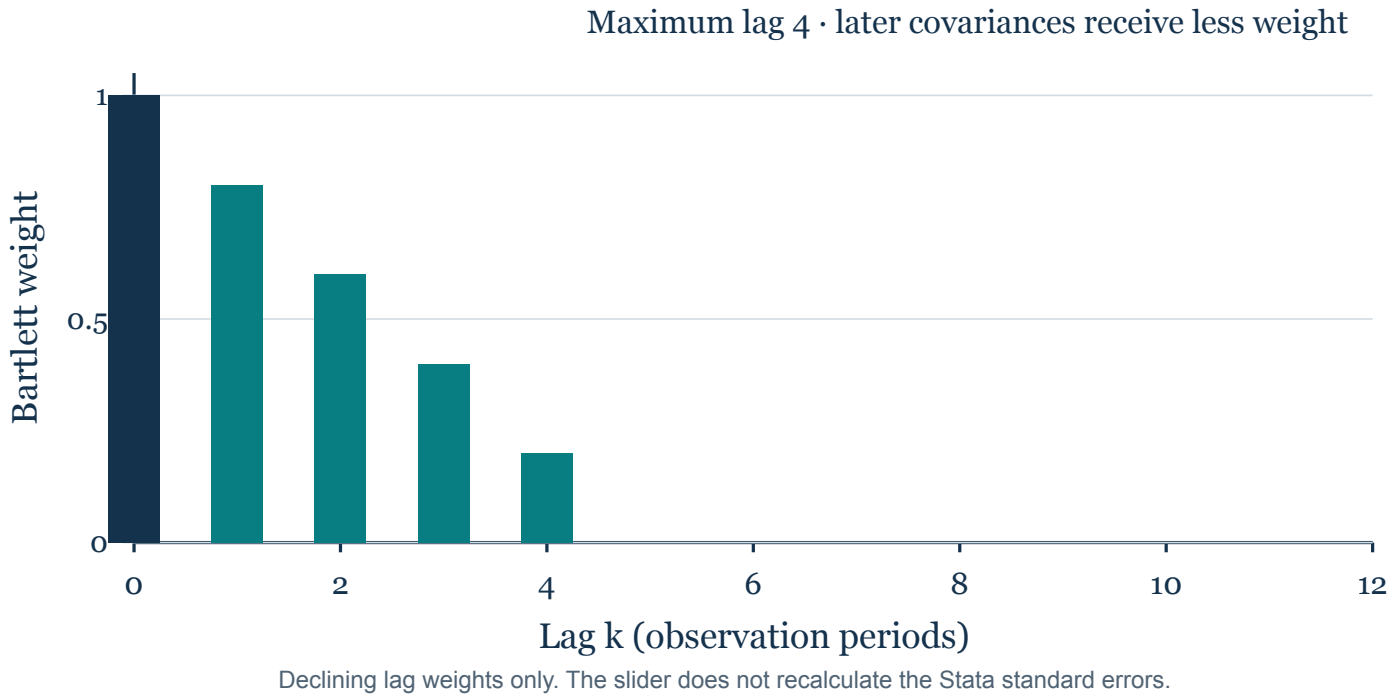
> $$\operatorname{Var}(\overline{u}) = \frac{\sum_t \operatorname{Var}(u_t) + 2\sum_{s<t} \operatorname{Cov}(u_s, u_t)}{n^2}$$

> Variance measures spread; covariance measures how two errors move together.

> Two unit-variance errors: covariance $0 \Rightarrow 0.5$; covariance $0.5 \Rightarrow 0.75$

# Use HAC to account for unequal spread and time dependence

Newey–West keeps the OLS coefficients and changes their estimated uncertainty.

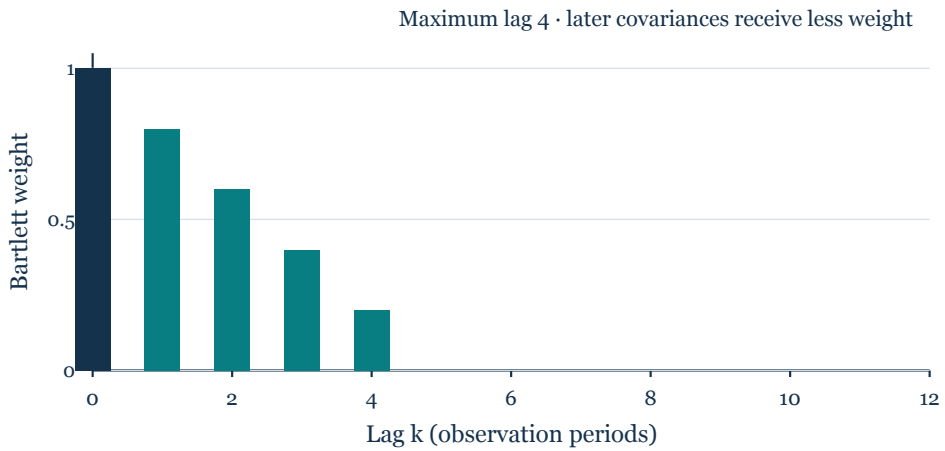



Declining lag weights only. The slider does not recalculate the Stata standard errors.

- $w_0 = 1, \quad w_k = 1 - k/(L+1), \quad k = 1, \ldots, L$
- For $L = 4$: $1, 0.8, 0.6, 0.4, 0.2$
- Weights apply to regressor–residual covariance contributions.

# Interpret the lag in calendar time

Interpret the selected maximum lag using months or quarters.




Declining lag weights only. The slider does not recalculate the Stata standard errors.

$$w_k = 1 - \frac{k}{L+1}, \quad k = 0, \ldots, L$$

Bartlett weights taper the contribution of later error covariances.

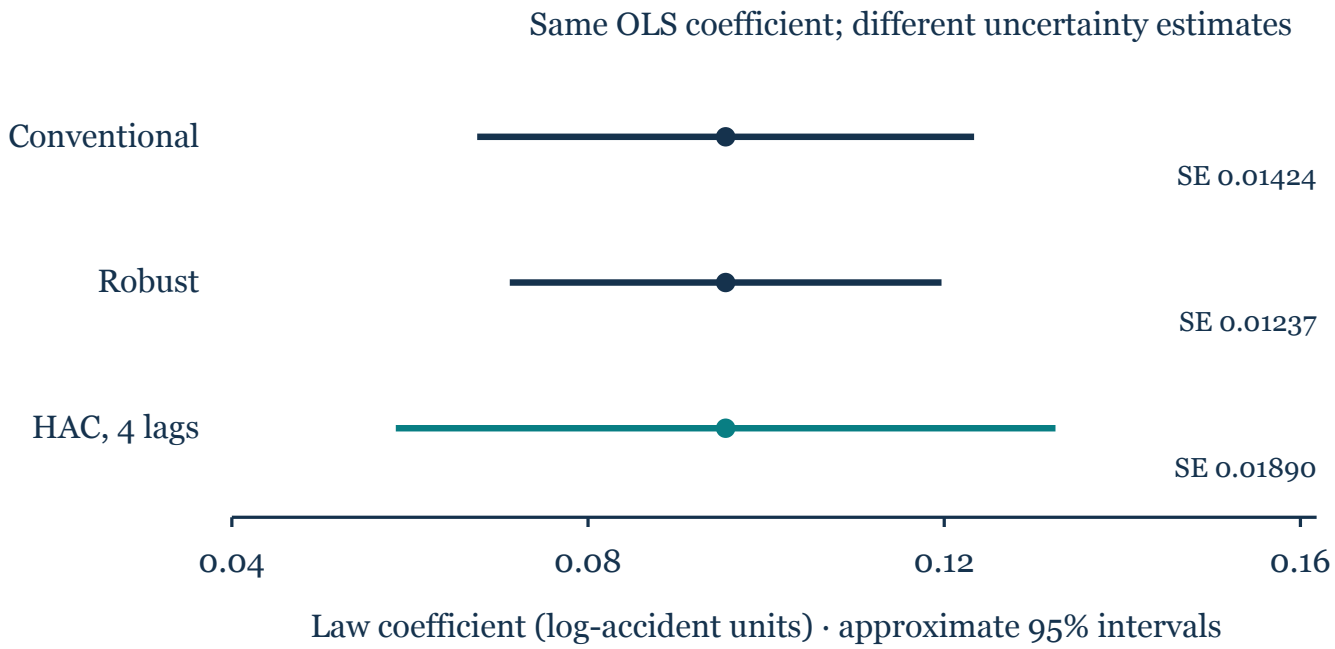
## Practice

For quarterly data, what calendar window is lag(12)? Does that option add twelve regressors?

**Explanation:** It reaches twelve quarters, or three years. It adjusts the covariance calculation; it adds no conditional-mean regressors.

# What stays fixed in the three fits?

Hold the fitted model fixed and compare its three SE calculations.




Actual Stata estimates and SEs · approximate intervals = b ± 1.96 × SE. Exact software intervals are in the output.

# What an SE adjustment cannot repair

HAC changes uncertainty; it does not remove a regressor's relation with the error.

**Practice**

A lagged outcome is correlated with the current error. What does an HAC option change, and what does it leave?

**Explanation:** It changes the estimated covariance around the fitted coefficients. It leaves the regressor–error correlation and potential coefficient inconsistency in place.

# Check whether past information is still missing

Dynamic completeness concerns whether the model uses the relevant past information.

> Dynamic completeness: additional relevant past information does not improve the model's conditional mean

> Adding a lag changes the fitted equation and may change the sample

> HAC addresses uncertainty from dependence that remains

# Choose the response that matches the problem

Adding lags, Newey–West and Prais–Winsten change different parts of the analysis.

> Missing mean dynamics → reconsider the equation.

> Valid mean model with suitable remaining dependence → justify covariance estimation.

> A structured error transformation is developed in the next meeting.

# Run the conventional regression

Find the law coefficient and understand the labels before comparing SEs.

```
. regress logacc trend i.month unem spdlaw beltlaw wkends

------------------------------------------------------------------------------
      logacc | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
       trend |   .0011011   .0002579     4.27   0.000     .0005889    .0016133
             |
        unem |  -.0212173   .0033974    -6.25   0.000    -.0279659   -.0144688
      spdlaw |  -.0537593   .0126036    -4.27   0.000    -.0787948   -.0287238
     beltlaw |   .0954526    .014235     6.71   0.000     .0671765    .1237288
      wkends |   .0033333   .0037761     0.88   0.380    -.0041675    .0108342
       _cons |   10.63986   .0630859   168.66   0.000     10.51455    10.76518
------------------------------------------------------------------------------
```

Calendar-indicator coefficient rows are omitted here. The output download contains the full numerical results.

# Run heteroskedasticity-robust inference

vce(robust) allows unequal error variance, while preserving the OLS coefficients.

```
. regress logacc trend i.month unem spdlaw beltlaw wkends, vce(robust)

------------------------------------------------------------------------------
             |               Robust
      logacc | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
       trend |   .0011011   .0002676     4.12   0.000     .0005696    .0016326
             |
        unem |  -.0212173   .0038238    -5.55   0.000    -.0288128   -.0136218
      spdlaw |  -.0537593   .0120059    -4.48   0.000    -.0776076    -.029911
     beltlaw |   .0954526   .0123682     7.72   0.000     .0708847    .1200206
      wkends |   .0033333   .0039443     0.85   0.400    -.0045016    .0111682
       _cons |   10.63986   .0681238   156.18   0.000     10.50454    10.77518
------------------------------------------------------------------------------
```

Calendar-indicator coefficient rows are omitted here. The output download contains the full numerical results.

# Run Newey–West on the same model

newey with lag(4) allows covariance contributions through four months.

```
. newey logacc trend i.month unem spdlaw beltlaw wkends, lag(4)

------------------------------------------------------------------------------
             |             Newey–West
      logacc | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
       trend |   .0011011   .0003897     2.83   0.006      .000327    .0018752
             |
        unem |  -.0212173   .0051302    -4.14   0.000    -.0314079   -.0110268
      spdlaw |  -.0537593   .0170449    -3.15   0.002    -.0876169   -.0199018
     beltlaw |   .0954526   .0188985     5.05   0.000     .0579131    .1329922
      wkends |   .0033333   .0028746     1.16   0.249    -.0023768    .0090434
       _cons |   10.63986   .0635823   167.34   0.000     10.51356    10.76616
------------------------------------------------------------------------------
```

Calendar-indicator coefficient rows are omitted here. The output download contains the full numerical results.

# Convert the fitted log difference into percent

Use 100[exp(b)−1] for the exact contrast between exponentiated fitted values.

```
. nlcom 100*(exp(_b[beltlaw])-1)

------------------------------------------------------------------------------
      logacc | Coefficient  Std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
       _nl_1 |   10.01567    2.07913     4.82   0.000     5.940653    14.09069
------------------------------------------------------------------------------
```

$g(b) = 100\{\exp(b) - 1\}$

$g'(b) = 100\exp(b) \approx 110.02$

$SE\{g(\hat{b})\} \approx |g'(\hat{b})| SE(\hat{b}) \approx 2.08$ percentage units

# Read the serial-correlation diagnostic

Breusch–Godfrey tests a stated no-serial-correlation null.

```
. estimates restore conventional
(results conventional are active now)

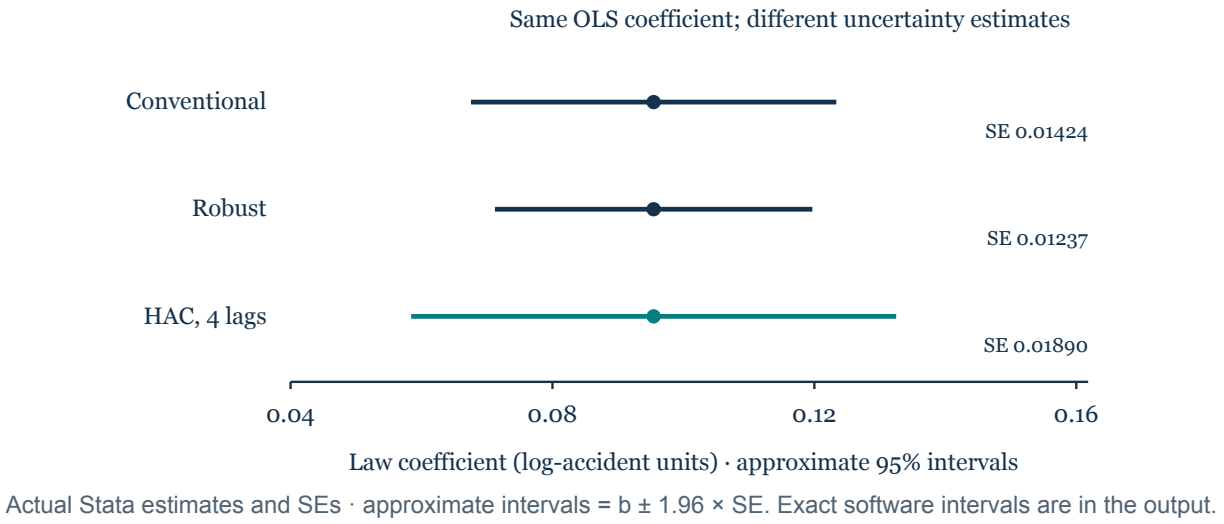
. estat bgodfrey, lags(1 4)

Breusch–Godfrey LM test for autocorrelation
------------------------------------------------------------------------------
    lags(p)  |          chi2               df                 Prob > chi2
-------------+----------------------------------------------------------------
       1     |         13.010               1                   0.0003
       4     |         16.907               4                   0.0020
------------------------------------------------------------------------------
                       H0: no serial correlation
```

- Supporting regression: current residual on original regressors and earlier residuals.
- Null: added residual-lag coefficients are jointly zero.
- Chi-square reference degrees of freedom: number of tested lag restrictions.

# Calculate, then explain the comparison

Calculate the test and interval, then report what they describe.




Actual Stata estimates and SEs · approximate intervals = b ± 1.96 × SE. Exact software intervals are in the output.

## Practice

Use b=0.09545264 and HAC4 SE=0.01889849 to calculate the approximate t ratio and 95% interval for b.

**Explanation:** t≈5.05. The large-sample interval is 0.09545264±1.96×0.01889849≈[0.05841, 0.13249], on the log-outcome coefficient scale. This is uncertainty about the fitted coefficient under the inference assumptions; it is not automatically an interval for a causal law effect.

# Apply the reasoning to quarterly inflation

Define the economic shock and time horizon before interpreting a future response.

> Inflation: rate of increase in the general price level

> Quarterly h = 4: the outcome one year ahead

> Local projection: a separate future-outcome regression for each horizon

> HAC lag 12 in quarterly data: a three-year covariance window

# Choose an action that addresses the actual problem

Connect the observed concern to the model, comparison or uncertainty calculation.

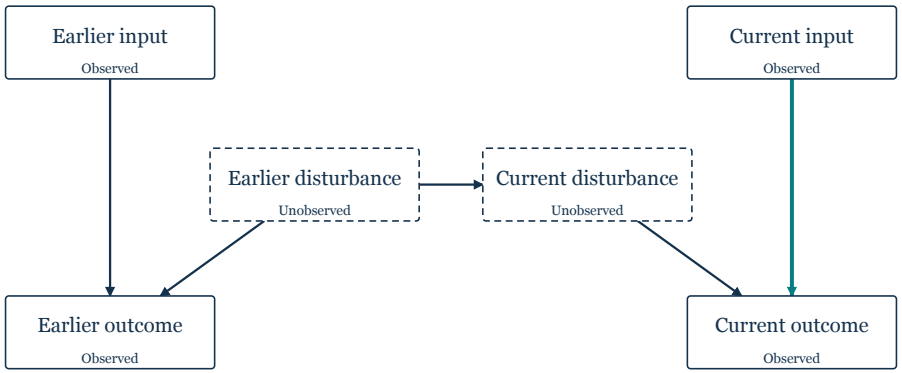
### Practice

A policy indicator responds to unobserved accident risk. Is the first remedy a longer HAC window?

**Explanation:** No. The named problem is policy endogeneity. A longer HAC window changes uncertainty estimates but does not create an exogenous policy comparison. Address the design or identify a defensible source of policy variation.

# DAG practice: persistent disturbances and HAC

Trace the routes before choosing a control.




1 open directed route · 0 open alternatives · gold: conditioned on

Assumed causal structure · dashed boxes: unobserved · gold boxes: conditioned on

**Practice**

The earlier disturbance affects the current disturbance, but the input nodes have no common-cause connection to these disturbances in the stated graph. Does disturbance persistence alone create regressor endogeneity? What does HAC change, and which arrows does it remove?

# DAG practice: persistent disturbances and HAC: explanation

Explain what conditioning changes in this stated graph.

> Persistence alone need not create endogeneity in this proposed model. It can create dependent outcomes and errors while the relevant input–disturbance mean conditions remain valid. Those are separate questions. HAC changes the estimated covariance of the coefficient under suitable exogeneity, dependence and sample conditions. It removes no causal arrows and keeps OLS coefficients fixed on an identical specification and sample. If the actual input selection process introduces a common-cause route to the disturbance, HAC does not close that new route.

# Choose an uncertainty statement for repeated months

Reproduce the evidence, then explain its economic meaning.

**Practice**

Run meeting_11.do. Confirm that conventional, robust and Newey–West fits retain the same law coefficient. Compare their SEs and approximate intervals. Explain why a smaller robust SE is possible, why four lags means four months here, and what none of these uncertainty options can establish about law adoption.

# Choose an uncertainty statement for repeated months: explanation

A reproducible result needs both a calculation and an interpretation.

> The common law coefficient is about 0.095453. The SEs are about 0.014235, 0.012368 and 0.018898 for conventional, robust and HAC4 inference. Using the common estimate plus or minus 1.96 times each SE gives different approximate intervals without changing the fitted model. A robust SE can be smaller because its covariance estimate responds to the actual relationship between regressor leverage and error spread. Four covariance lags are four months in this monthly dataset; the option does not add four regressors. HAC allows a specified treatment of remaining time dependence under suitable conditions, while none of these choices makes law adoption exogenous or removes omitted dynamics from the conditional mean.

# Exit check: connect the model, estimate and SE

Explain the three choices as responses to different empirical concerns.

> Same equation/sample → same OLS coefficient.

> Different justified covariance calculations → potentially different intervals.

> A policy effect still needs an assignment argument.

---

**Practice**

In one paragraph, compare adding a lag, using Newey–West and fitting Prais–Winsten. Interpret lag(4) for this dataset.

**Explanation:** Adding a lag changes the mean equation and may change the sample and coefficients. Newey–West keeps OLS coefficients on the same equation and sample, while changing their estimated covariance; lag(4) here reaches four months. Prais–Winsten uses an AR(1) error model to transform the fit and may change coefficients and SEs. None of these changes, by itself, creates an exogenous policy comparison.